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Ranveer Thakur Chand:

This lab was on Exploratory Data Analysis, basically just exploring a dataset before jumping into any modeling. We used the Titanic dataset and went through statistics and visualizations to find patterns.

The first issue I ran into was that all the code in every cell was on one line, so nothing would run. I had to go through each cell and reformat the code before I could do anything.

Once that was fixed, I loaded the dataset with pandas and ran `.info()` and `.describe()` to get a basic picture of the data missing values, average age, survival rate, stuff like that. Then we got into visualizations using Seaborn and Matplotlib. Seaborn is basically built on top of Matplotlib and makes plotting a lot simpler. I ran countplots for survival by class and gender and used a FacetGrid with histograms to compare age distributions between survivors and non-survivors.

For the experiments I made a countplot for port of embarkation and then did a boxplot and violin plot comparing fares between survivors and non-survivors, which showed survivors tended to pay higher fares. The big takeaway was that visualizing data shows things raw numbers can't, like outliers, skewed distributions, and relationships between variables that you'd miss just looking at a mean or count.

John Nasir:

Exploratory Data Analysis (EDA) is described as the detective work of data science. Before running this lab, my understanding of data analysis was the focus on the final models. However, this lab showed me that a model is only as good as my understanding of the data built into it.

Using the historical Titanic passenger dataset, this lab showed how to explore a raw dataset, spot problems, find missing data, and pull out useful insights before doing any machine learning.

The first phase of the lab focused on was desc. statistics using automatted tools like `df.info()` and `df.describe()`. While these provided a necessary overview such as establishing an overall survival rate of approximately 38% and showing missing data fields in the 'Age' and 'Cabin' columns the insights emerged during the visualization phase.

Using Seaborn and Matplotlib really helped turn the raw Titanic numbers into visuals that made the social patterns jump out immediately.

When I compared survival rates across passenger classes, it became obvious that first-class passengers had a much better chance of surviving than those in third class. That showed how much someone's social and economic status affected their outcome.

The gender and age plots told gave me a clear story. The countplots showed a huge survival advantage for women, which lines up with the "women and children first". When I used a FacetGrid to look at age, I noticed a survival spike for very young kids something you'd never notice by just looking at averages.

During the experimentation part, I also looked at where passengers (Embarked) and how fares

were distributed. The fare boxplots showed that people who paid more generally had higher survival rates, which matched the class findings. The embarkation plots suggested that different ports had different passenger demographics which added more context to the dataset.

Overall, this lab changed how I think about data visualization. I used to see charts as something you make to present results. Now I know them as tools for understanding the data itself spotting missing values, weird patterns, outliers, and relationships. Averages can hide important shapes in the data, but histograms and boxplots reveal them.

The skills from this lab handling missing data, visualizing categories, and checking continuous variables for skew are useful in any field. Whether it's customer churn, patient health data, or fraud detection, good exploratory data analysis is the first step in turning messy data into something.

Trevor Oakum:

Working through this lab, I learned how important Machine Learning can be when quantifying large data sets, and how crucial it is that these models learn and understand the data being presented to them as much as possible. Speaking of the data, the goal of Exploratory Data Analysis, particularly in this situation, is to analyze large sets of data that are uniquely different from one another, and combine them to tell necessary information based on the scene where the data was collected.

With this scenario (data similar to the Titanic), the most likely scenario for who would be survivors would be women, particularly anyone below the age of 25, and those of higher income. With this data, we can understand not only why this class would most likely survive, but also the importance of visualizing data. When data becomes visualized, you can see clusters. Compared to a mean or an average, it's important to find these clusters because these clusters, in this scenario, would tell us larger groups of members based on particular factors that would account for rates. One could then extract averages from these clusters and see the survival rate amongst these clusters.

Now, with the visualized data in front of us, I'll explain why it's important to also draw conclusions from this data, as this data "tells a story" and we can make inferences to understand how to better prepare for the future. Let's break down the demographics and understand what we can determine.

In the case of natural disasters, young adult men (18-35) would be more likely to be left behind due to the idea of evacuating those who may not be able to survive on their own.

Those who were older by generation, up to those who require assistance, would get priority as they reach more mature ages. With those being older, regardless of gender, it would be difficult to move them to safer crafts due to the speed and assistance required. With women and children, they would be given priority over all else due to the perceived nature at the time that they would both have a difficult time fending for themselves, and that they

must be the ones to tell the story. Finally, with income, this reason is simple. On traditional large ships, the more expensive suites would be built higher up (allowing for a higher chance of accessing lifeboats), and the fact that when the Titanic hit, the place that was immediately struck was the bottom, allowing for more time for the wealthy individuals to evacuate quicker, while the lower cabins (where the cheaper rooms were), flooded first due to impact). Overall, this is a very surface-level interpretation of this data, but it's important to not only visualize this data, but make important inferences of this data, so that not only can one understand what the graph is saying, but in the case of training algorithms, have them understand the data, and why the data is telling us that so we can strive for more accurate models while trying to eliminate issues commonly seen in statistics such as bias.

Freddy Rodriguez:

My biggest "Aha!" moment was to realize that data has a shape and tells a story. A spreadsheet of 891 passengers looks like a meaningless grid. But the moment I plotted passenger classes (Pclass) and gender (Sex) against who survived (Survived), the data came alive. I mathematically proved history: first-class passengers and women had a vastly higher survival rate. Charts like bar graphs and boxplots expose the exact clues a computer needs to make accurate predictions.

Challenges Faced and Problem-Solving

As I made my own charts, I ran into two technical roadblocks that forced me to solve a problem:

The Messy Chart Problem: Mapping ticket prices using a standard bar chart created a chaotic, unreadable graph because there were hundreds of different prices. My Strategy: I switched to a boxplot

to group the continuous numbers into a clean summary showing the middle value (median) and overall spread.

The Outlier Problem: A few wealthy passengers paid massive fares over \$500, stretching our chart's axis sky-high and squishing the other 95% of data. My Strategy: I used a line of code (`plt.ylim(0, 300)`) to temporarily crop out extreme outliers and bring the main crowd into sharp focus.

Connections, Questions, and Changing Our Mindset

This lab connected basic math to a major data science rule: graphs reveal truths that raw numbers hide. A standard average can be completely warped by a single massive outlier. Visualizing the data showed us the true distribution and validated the rule of "women and children first". This exploration sparked me deep questions:

Did the port where people boarded (Embarked) actually affect survival, or did certain ports just have more rich, first-class passengers board there?

Missing Pieces: The data is missing 177 passenger ages. Should I fill those blanks with a basic average, or guess their age smarter based on their ticket class?

Ultimately, my mindset shifted from "Model-First" to "Data-First." I used to think data science was just about using complex algorithms. Now I know a model is only as good as the data you feed it. Because far more people died than survived, the data is "imbalanced". A computer could blindly guess "everyone died" and be 61% accurate, but it would be totally useless. This lab taught me to look at the graphs, clean the data, and understand the human story behind the numbers first.